Scalable and Flexible Network Functions using SDN White-Boxes

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UNSW AUSTRALIA

Never Stand Still

DATA 61

ANZ SDN

AUSTRALIA & NEW ZEALAND SOFTWARE DEFINED NETWORKING

Who are we and why are we here?

- Vijay: PhD @ UCLA/Bell-Labs; Switch Vendor @ silicon-valley start-up
- Craig: PhD @ Macquarie Uni; Network Operator @ PowerTel & CSIRO

- We are excited by innovation, but disappointed by research ⟵⟶ practice gap
  - Depth vs. breadth; optimal vs. simple; innovative vs. operable

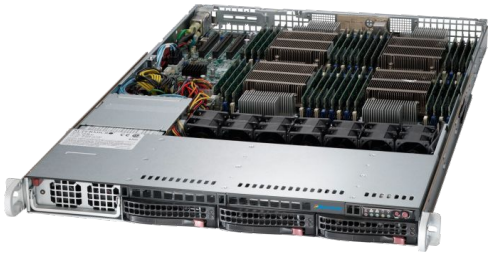
- Software Defined Networking (SDN): a fresh approach?
  - Software "brain" separated from hardware "brawn"
  - Opportunity: Innovate at software velocity, apply at hardware data-rates
  - Challenge: software providers decoupled from hardware (white-box) vendors
  - ANZ-SDN Alliance: community to foster SDN ecosystem

- Focus: Network Functions (NFs) for classification & cyber-security

Network Functions (NFs)

- Special-purpose operations on packets
  - Service Providers: DPI, load-balancer, …
  - Enterprises: Firewalls, NAT, …
- Physical appliances (middle-boxes) are expensive
  - Custom ASICs / FPGAs
- Virtual appliances (VNFs on x86) difficult to scale
  - General-purpose CPUs not optimized for network operations

How can SDN Help?



Hardware: dumb+fast on "flows"

- Abstraction: match + action + counter
- Match: any header field(s)
- Action: out-port, mirror, drop, edit

- Multi-vendor "white-box"
- Scalable: 400 Gbps – 6.4 Tbps
  - 1m+ flow table entries (FIB)
- Cost-effective: $20-40K

Software: smart+slow on "packets"

- Deep inspection / stateful operations
- Intelligence (e.g. optimization, ML/AI)

- Commodity servers cheap
  - Elastic scale-out, DPDK support
  - Flexible and agile
  - Software dev environments / tools

So how to put the two together?

- Operate on flows when you can and packets when you must
- 75-80% of network traffic is carried in top 5% ("long") flows
  - At UNSW: ~300K concurrent flows at peak hour; ~6k flows transfer > 4MB [1]
  - Corroborates with prior mice/elephant studies in enterprise and carrier networks [2]
- Here's a simple idea:
  - Operate on mice packets in software (VNF)
  - Offload elephant flows to hardware (SDN)
- Potential benefits of "SDN Accelerated NF" (SANF):
  - Software VNF load reduces by 4x → lower cost
  - Hardware flow table (FIB) is small → better scalability

[1] S. Madanapalli, M. Lyu, H. Kumar, H. Habibi Gharakheili, V. Sivaraman, "Real-time Detection, Isolation and Monitoring of Elephant Flows using Commodity SDN System", IEEE/IFIP NOMS workshop on Experience Sessions, Taipei, Taiwan, Apr 2018.

[2] T. Mori et al, "Identifying elephant flows through periodically sampled packets", ACM IMC, Taormina, Sicily, Italy, 2004.

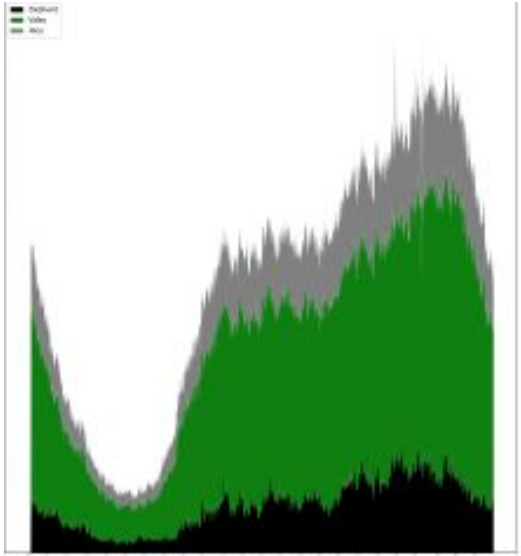
Example Uses

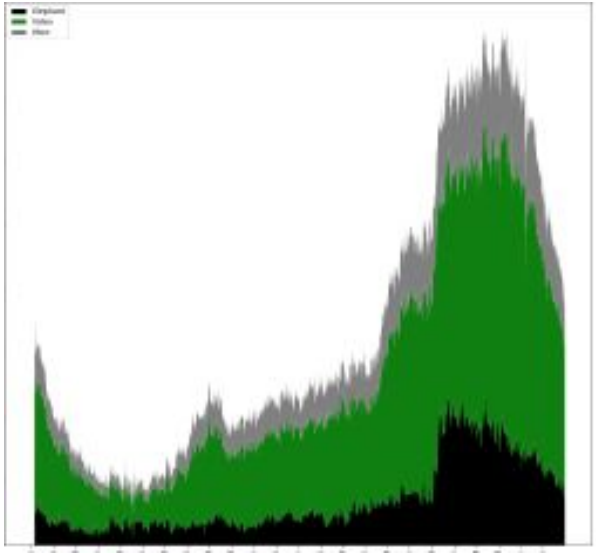
- Example 1: Load-Balancer
  - Handle first packet of flow in VNF, then insert flow table rule to send to selected outport
- Example 2: NAT
  - Determine port mappings in VNF based on first packet of flow; subsequent packets in h/w
- Example 3: Traffic classifier
  - Keep state on flow volume in VNF; Insert flow table rule in h/w on reaching threshold
  - Extract flow telemetry and classify based on behavioral profile
- Example 4: Stateful firewall
  - Determine state on outbound flow based on first packet; insert flow rule for return traffic
- These examples and their complexity/scalability modeled in [3]

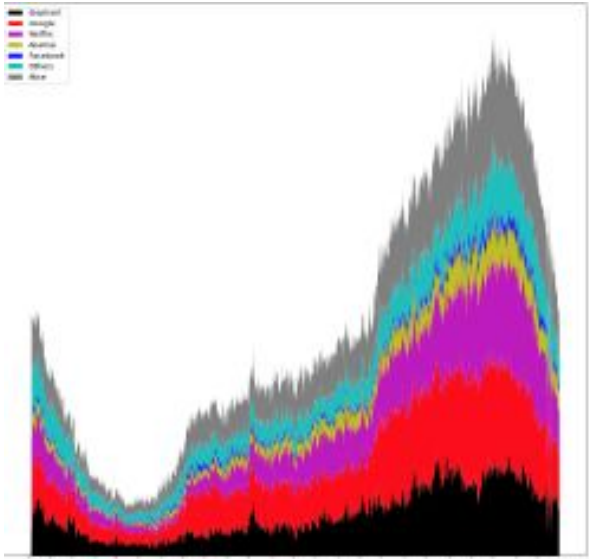
[3] V. Sivaraman, H. Habibi Gharakheili, M. Lyu, H. Kumar, C. Russell, "Scaling Network Functions via SDN Acceleration", under review with IEEE Journal on Selected Areas in Communications (JSAC), 2018.

NF Use-Case 1: Traffic Classification

- Motivation: Understand broadband usage pattern
  - Break-down of browsing, video, and download traffic over a day
  - Composition of video traffic by provider (Youtube, Netflix, …)

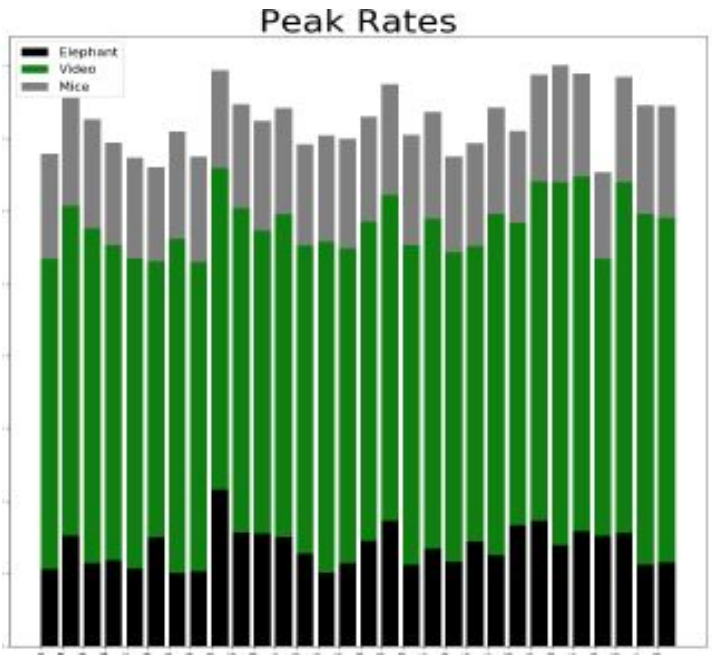


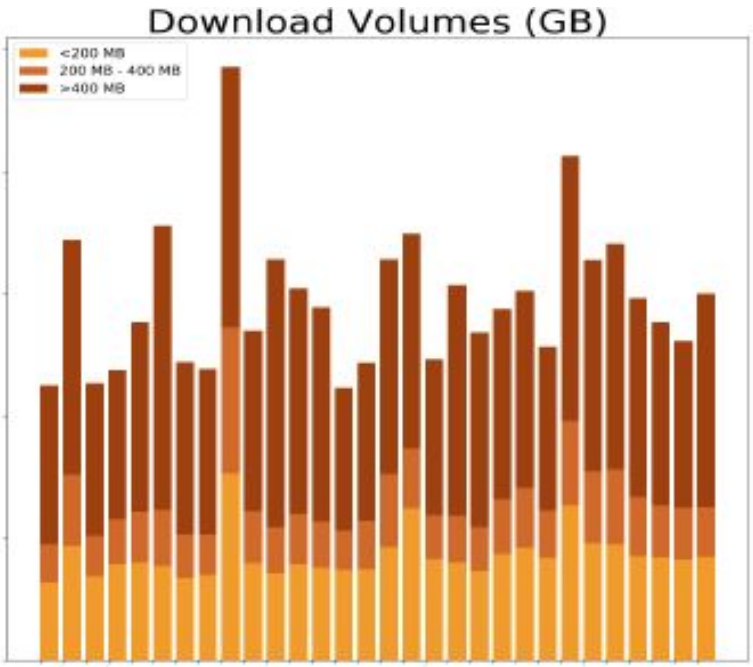




Traffic Classification (contd.)

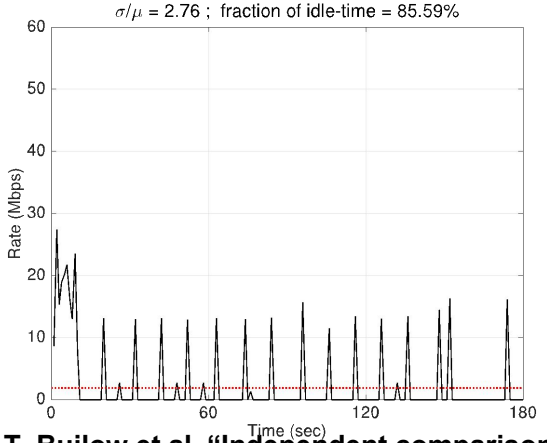
- Motivation: Manage demand variability (peak provisioning)

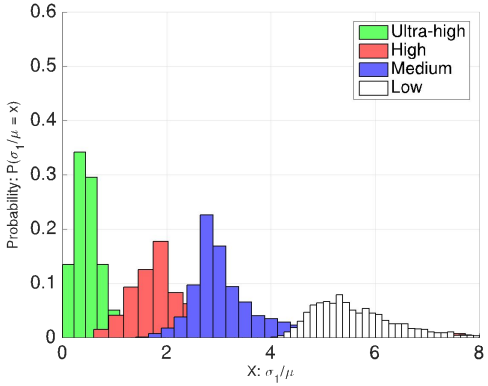


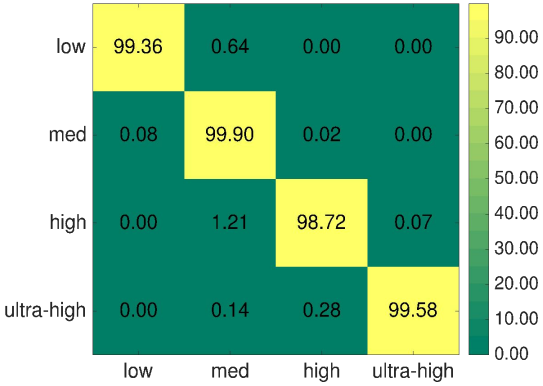


DPI vs Flow Classification

- Approach: Classify long flows based on behavioral profile [4],[5]
  - No inspection of packets (except DNS)
  - VNF tracks “short” flows, offloads “long” flows to SDN white-box hardware
  - Extracts statistical profile of long flows and does machine learning (ML) classification





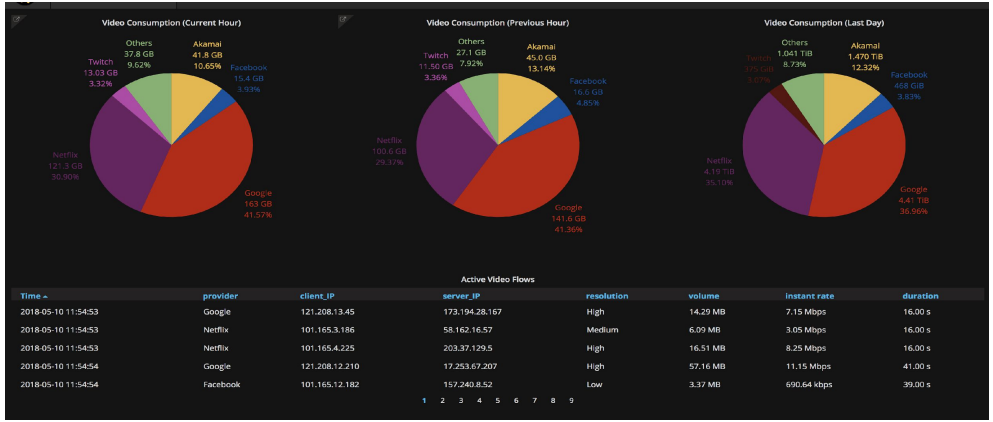


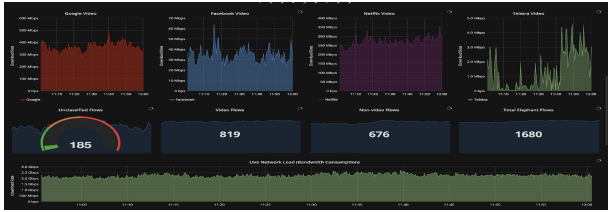
[4] T. Bujlow et al, “Independent comparison of popular DPI tools for traffic classification”, Computer Networks, 76(c):75 – 89, Jan 2015.

[5] A. Dainotti et al., “Issues and future directions in traffic classification”, IEEE Network, 26(1): 35–40, Jan 2012.

TeleScope: Flow Classification

- Fully implemented and operational today [6]
  - Uses NoviFlow SDN switch, Ryu Controller, NFF-GO DPDK, TensorFlow ML, …
  - Real-time UI, logging of every long flow, analysis/insights each midnight
  - Operating on 10 Gbps mirror traffic at UNSW for past 12 months





[6] H. Habibi Gharakheili, M. Lyu, Y. Wang, H. Kumar, V. Sivaraman, "iTeleScope: Intelligent Video Telemetry and Classification in Real-time using Software Defined Networking", under review with IEEE/ACM Transactions on Networking, 2018.

TeleScope: Commercial offering

- IP licensed to Canopus Networks
  - In commercial trials with Telcos
  - Showing insights into video consumption (real-time / hourly / daily)
    - Identifies 400+ video content providers, including rate, resolution, duration
  - Showing insights into large downloads (real-time / hourly daily)
    - Storage, CDNs, games (Fortnite), …
  - Developing signatures for interactive video, streaming music, gaming, VR, …
- Partnering with white-box BNG vendors for in-line b/w management
  - Framework for <u>customer experience optimization</u> based on theory of "marginal utility"
  - Open, flexible, and verifiable framework that addresses net neutrality concerns [7]

[7] V. Sivaraman, H. Habibi Gharakheili, S. Madanapalli, H. Kumar, "Competing on User Experience in a Post-Neutral World", ACM Symposium on SDN Research (SOSR), ONS Santa Clara, Apr 2019.

NF Use-Case 2: Cyber-Security

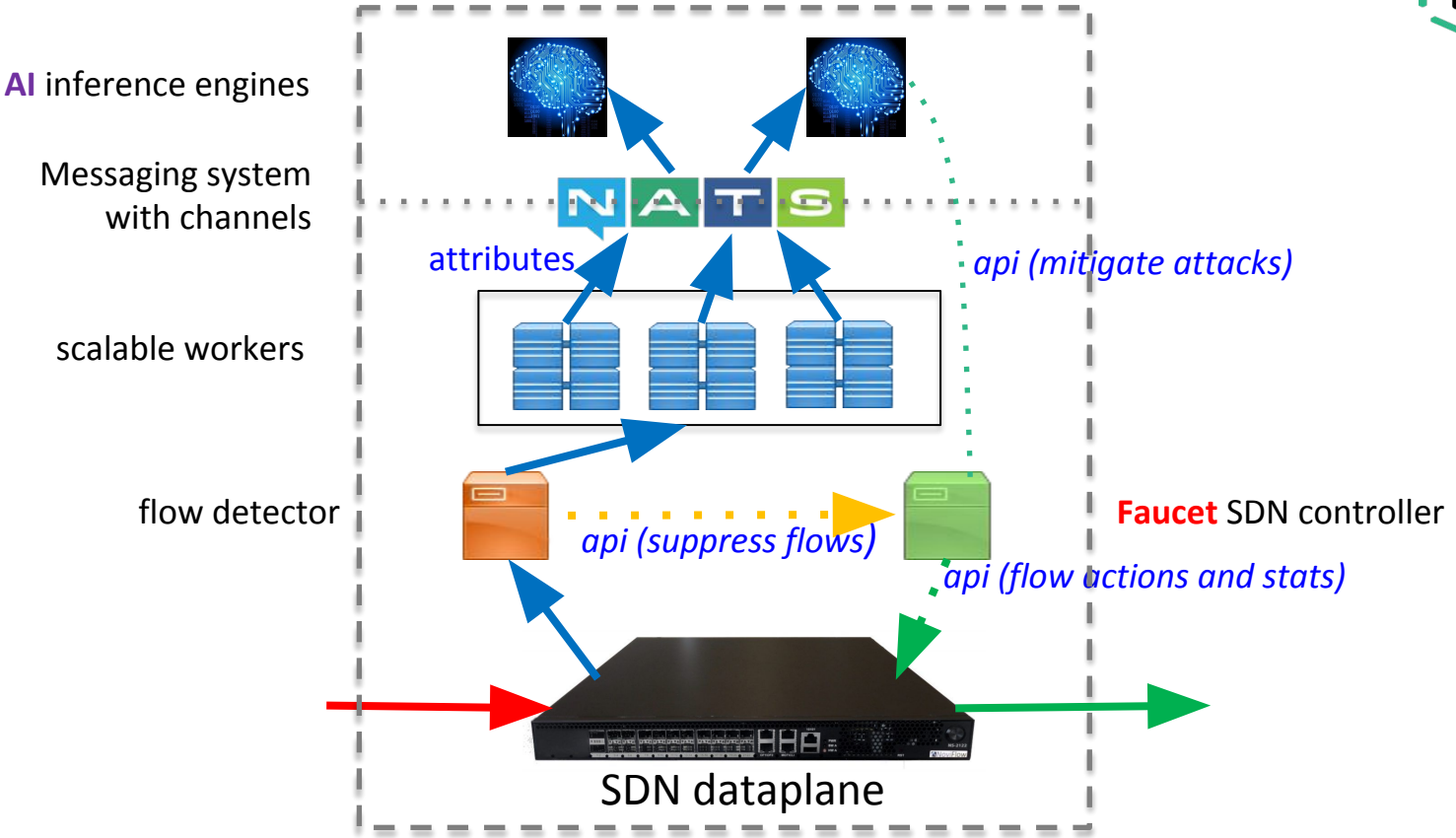
- Appliance based security solutions are:
  - Expensive → 10G/100G service may be cheaper than the security appliance!
  - Bundled → can't combine "best-of-breed" or get "second opinion"
  - Performance-limited → features have to be turned off for line-rate performance
- Our goal: build platform (**Nozzle**) to augment current solutions:
  - Leverages white-box SDN hardware and commodity server-class compute
  - is scalable (modular expansion of capacity as needed)
  - is flexible (best-of-breed components)
  - is low-cost (commodity components)
    - set of "**workers**" extract attributes ← **SDN**
    - set of "**inference engines**" make deductions ← **AI**

Why use SDN for cyber-security?

- The "***software defined***" ecosystem gives us the ability to do a lot more than shunt packets from A to B

- With **scalable, server-class compute** and a **programmable control plane** we can
  - extract flow attributes
  - perform complex data analytics on flows in near real-time
  - use ML and AI techniques
  - "close-the-loop" and control packet forwarding based on the analytics (which is ideal for cyber-security applications)

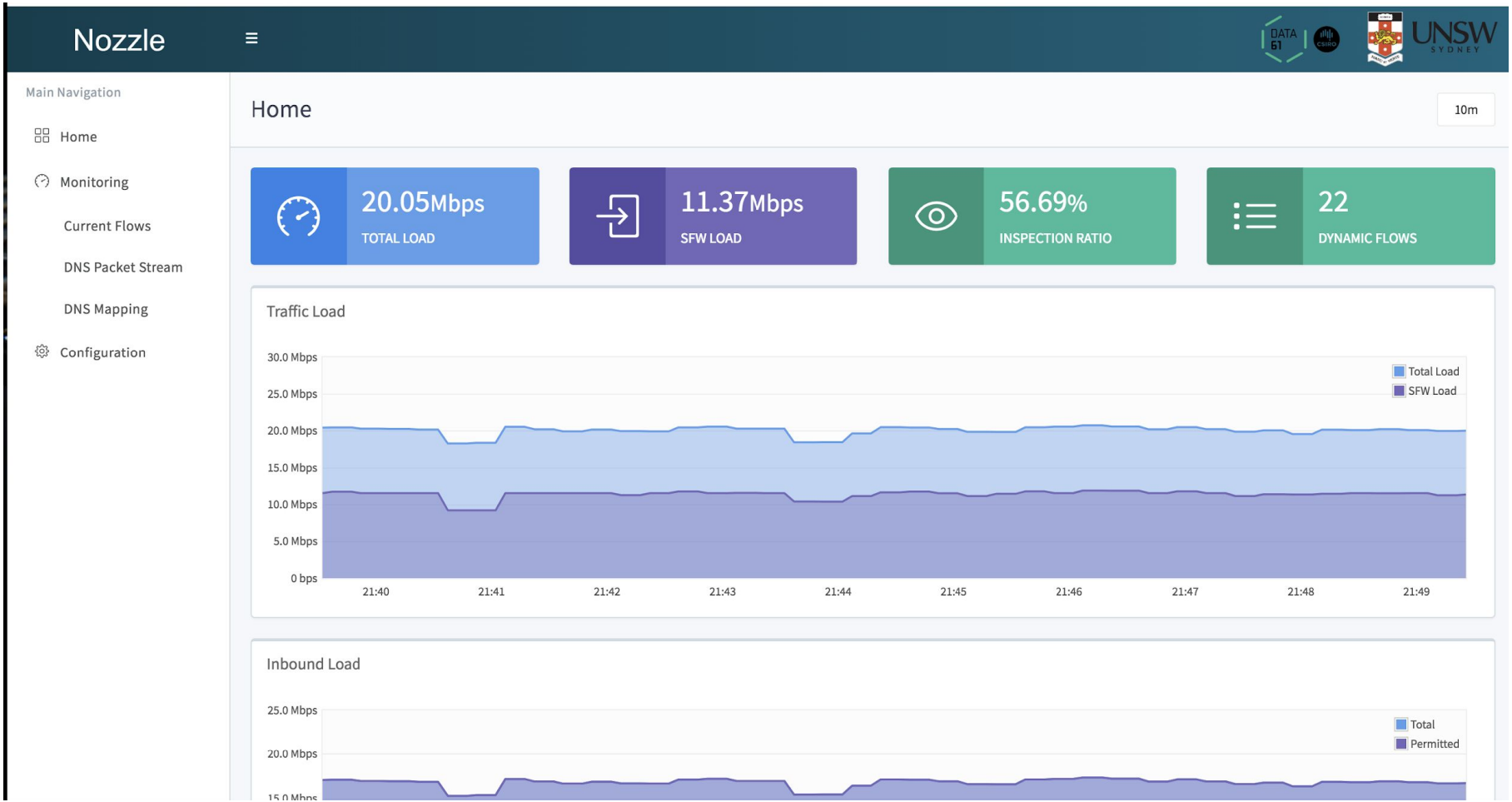
What Nozzle looks like

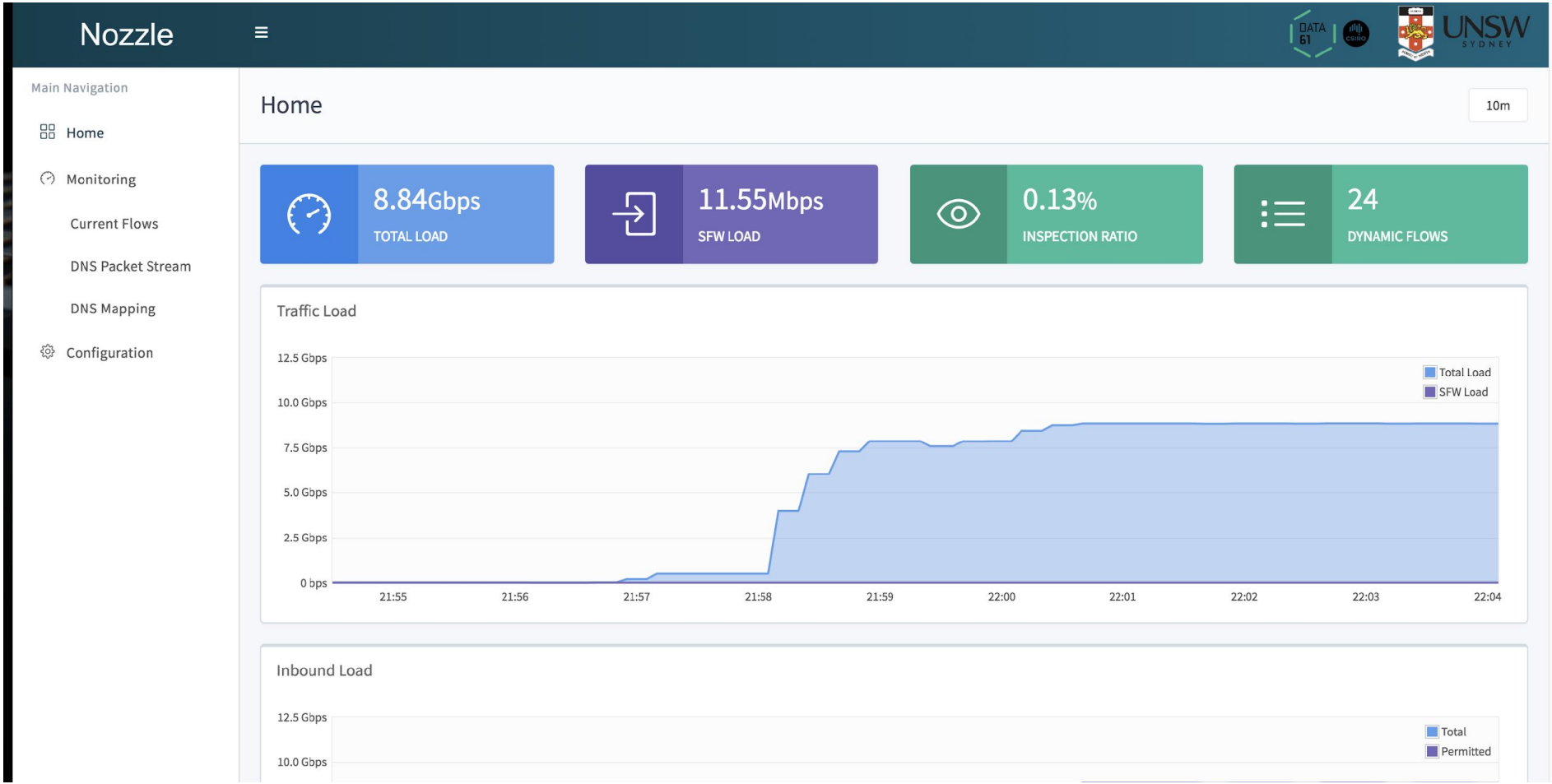


Nozzle technology

- **Hardware**
  - SDN switch (OpenFlow1.3 with multi-tables and 100Gb/s interfaces, NoviFlow in our case)
  - Compute servers (some equipped with 10Gb/s Intel DPDK capable NICs)

- **Software**
  - Ryu SDN controller application (Python)
  - flow extraction module (Go)
  - NATS for pub/sub messaging utilizing Google Protocol Buffers
  - ML algorithms for security analytics (Python and Go)
  - Time series databases (Prometheus and ElasticSearch)
  - Visualisation and Operator UI (ReactJS)


Nozzle
Main Navigation
Home
Monitoring
Current Flows
DNS Packet Stream
DNS Mapping
Configuration
Home
10m
20.05Mbps
TOTAL LOAD
11.37Mbps
SFW LOAD
56.69%
INSPECTION RATIO
22
DYNAMIC FLOWS
Traffic Load
30.0 Mbps
25.0 Mbps
20.0 Mbps
15.0 Mbps
10.0 Mbps
5.0 Mbps
0 bps
21:40
21:41
21:42
21:43
21:44
21:45
21:46
21:47
21:48
21:49
Total Load
SFW Load
Inbound Load
25.0 Mbps
20.0 Mbps
Total
Permitted


Nozzle
Main Navigation
Home
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Home
10m
8.84Gbps
TOTAL LOAD
11.55Mbps
SFW LOAD
0.13%
INSPECTION RATIO
24
DYNAMIC FLOWS
Traffic Load
12.5 Gbps
10.0 Gbps
7.5 Gbps
5.0 Gbps
2.5 Gbps
0 bps
21:55
21:56
21:57
21:58
21:59
22:00
22:01
22:02
22:03
22:04
Total Load
SFW Load
Inbound Load
12.5 Gbps
10.0 Gbps
Total
Permitted

Nozzle

Main Navigation

- Home
- Monitoring
- Configuration
  - Static ACL's
  - System Variables

ACL Configuration

Discard Changes | Save Configuration

Inbound ACL

EtherType	Src Address	Dst Address	Protocol	Src Port	Dst Port	Action	
IPv4	Any	202.9.2.1	TCP	Any	22	ALLOW	
IPv4	Any	202.9.2.1	TCP	Any	80	ALLOW	
							Add Rule

Outbound ACL

EtherType	Src Address	Dst Address	Protocol	Src Port	Dst Port	Action	
							Add Rule

Nozzle

Main Navigation

- Home
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  - DNS Packet Stream
  - DNS Mapping
- Configuration

Current Flows

Table 0

Select Columns

Flow Metadata			Match Fields									
priority	packet_count	byte_count	eth_type	ip_src	ip_dst	ip_proto	tp_src	tp_dst	icmpv4_type	icmpv6_type	in_port	Actions
32768	2893	484176	0x800	1.2.3.100	5.6.7.100	6	63047	80			3	OUTPUT:4
32768	2760	2337575	0x800	5.6.7.100	1.2.3.100	6	80	63047			4	OUTPUT:3
32768	3439	567056	0x800	1.2.3.100	5.6.7.100	6	62471	80			3	OUTPUT:4
32768	3228	2711092	0x800	5.6.7.100	1.2.3.100	6	80	62471			4	OUTPUT:3
32768	3131	2717052	0x800	5.6.7.100	1.2.3.100	6	80	55337			4	OUTPUT:3
32768	3292	558152	0x800	1.2.3.100	5.6.7.100	6	55337	80			3	OUTPUT:4
32768	3133	2713679	0x800	5.6.7.100	1.2.3.100	6	80	54712			4	OUTPUT:3
32768	3310	558950	0x800	1.2.3.100	5.6.7.100	6	54712	80			3	OUTPUT:4
32768	3174	550200	0x800	1.2.3.100	5.6.7.100	6	57135	80			3	OUTPUT:4
32768	3067	2715103	0x800	5.6.7.100	1.2.3.100	6	80	57135			4	OUTPUT:3

Security Analysis "*attributes*"

- **DNS attributes**
  - query & response volume, solicited vs. unsolicited
  - #RRs, TTL, sub-domain length, udp payload size, …
- **Host attributes**
  - #TCP-connections, volume, connection-rate, …
- **Flow attributes**
  - long-flow data volume, 5-tuple end-points, SSL/TLS cipher-suites, …
- **Intra-flow profile**
  - inter-packet gaps, packet size distribution, …
- **Other signaling attributes**
  - NTP, DHCP, SNMP, SSDP, SIP, etc.

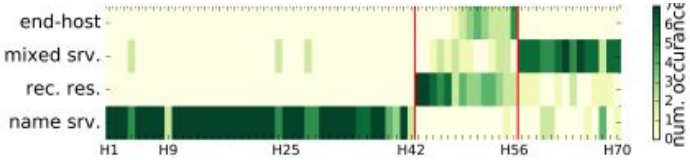
A deeper look at the DNS Attribute

- DNS:
  - comprises < 0.1% of traffic volume; often allowed through firewalls
  - wealth of information on external services accessed & internal servers
  - used for DoS, data exfiltration, reflection/amplification attacks

- DNS worker in **Nozzle**
  - SDN flow-rule + Intel DPDK with Golang
  - publishes aggregated DNS records to NATS every second

Analysis of UNSW's DNS Traffic

- Less than 10 Mbps (port 53)
- Extracted 4 attributes per-host:
  - Outbound queries / inbound queries
  - # external servers, # external clients, fraction of active time
- Built simple clustering-based ML classifier:
  - UNSW: 40 name servers; 14 resolvers; 368 public-servers



- Result: automated, dynamic, <u>daily</u> inventory of key assets
  - DNS resolvers, DNS name servers, external-facing web-servers

[8] M. Lyu, H. Habibi Gharakheili, C. Russell and V. Sivaraman, "Mapping an Enterprise Network By Analysing DNS Traffic", Passive and Active Measurements (PAM) Conference, Puerto Varas, Chile, Mar 2019.

DNS Attacks and Threats

- Evidence of attack vectors (what and how):
  - DNS scan of network
  - Recursive resolvers used for reflection attacks
  - Public-server DoS-sed with DNS responses

- DNS exfiltration possible and happening
- ML built using attributes: [9]
  - length, entropy, numerical, capital, dots, longest label, reputation
- Our machine detects exfiltration (e.g. cnr.io) in <u>real-time</u>

[9] J. Ahmed, Q. Raza, H. Habibi Gharakheili, C. Russell, V. Sivaraman, "Real-Time Detection of DNS Exfiltration and Tunnelling from Enterprise Networks", IFIP/IEEE International Symposium on Integrated Network Management (IM), Washington DC, USA, Apr 2019.

Nozzle

Main Navigation

- Home
- Monitoring
  - Current Flows
  - DNS Packet Stream
  - DNS Mapping
- Configuration

DNS Mapping

2019-01-28

Summary | DNS Servers | DNS Caches | End Hosts | Requested Hosts

Total DNS Hosts Classified	7
DNS Servers	3
DNS Caches	2
End Hosts	2
Requested Hosts	93

Nozzle

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DNS Mapping

2019-01-28

Summary | DNS Servers | DNS Caches | End Hosts | Requested Hosts

Address	Name	Type	Received Query Count	Sent Response Count	Number of Clients Served
202.9.4.227	ns2.centie.net.au	Authoritative	2023	2021	482
202.9.4.226	ns1.centie.net.au	Authoritative	1954	1953	472
202.9.4.229	ns1.nozzle.centie.net.au	Forwarding	8	8	7

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DNS Mapping

2019-01-28

Summary | DNS Servers | DNS Caches | End Hosts | Requested Hosts

Address	Name	Sent Query Count	Received Response Count	Number of Servers Queried
202.9.4.103	marshall.centie.net.au	7154	7157	191
202.9.4.118	drz.centie.net.au	3597	3604	128
2405:b000:a80:64::4:103	marshall.centie.net.au	1524	1528	145
2405:b000:a80:64::4:118	drz.centie.net.au	818	815	112

Nozzle

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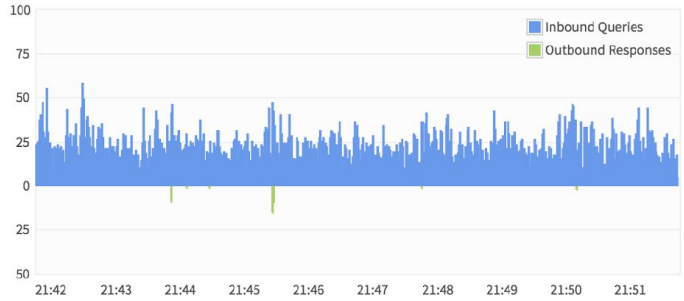
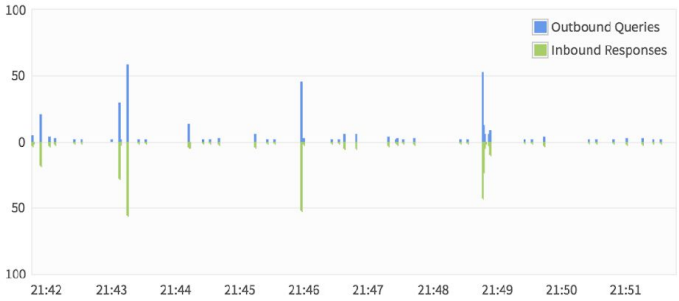
DNS Packet Stream

10m

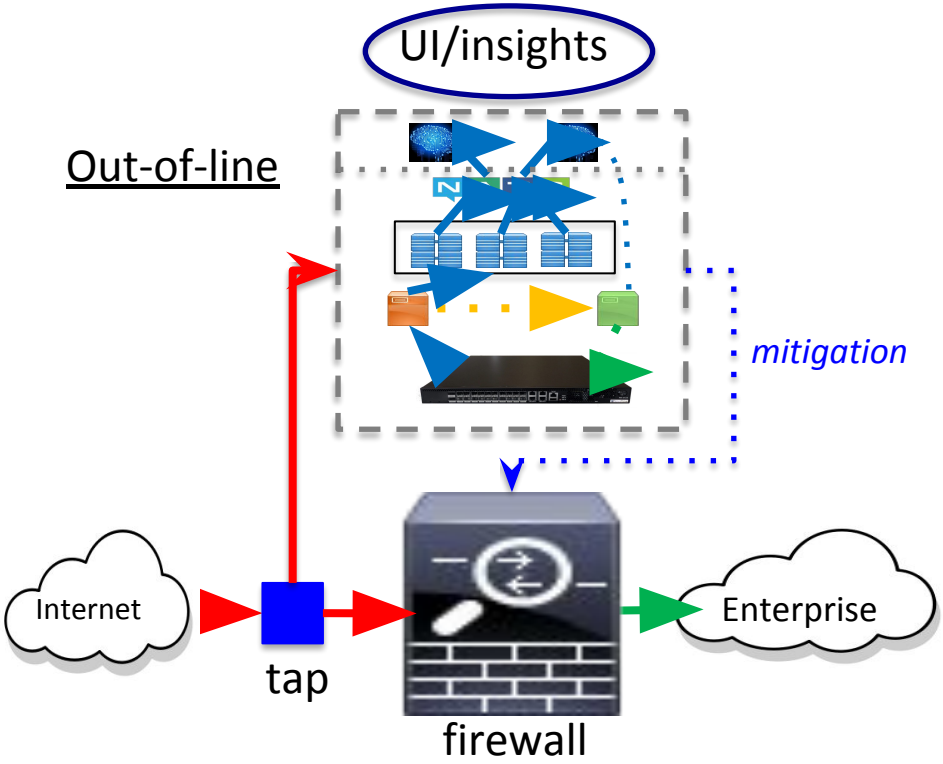
Show All

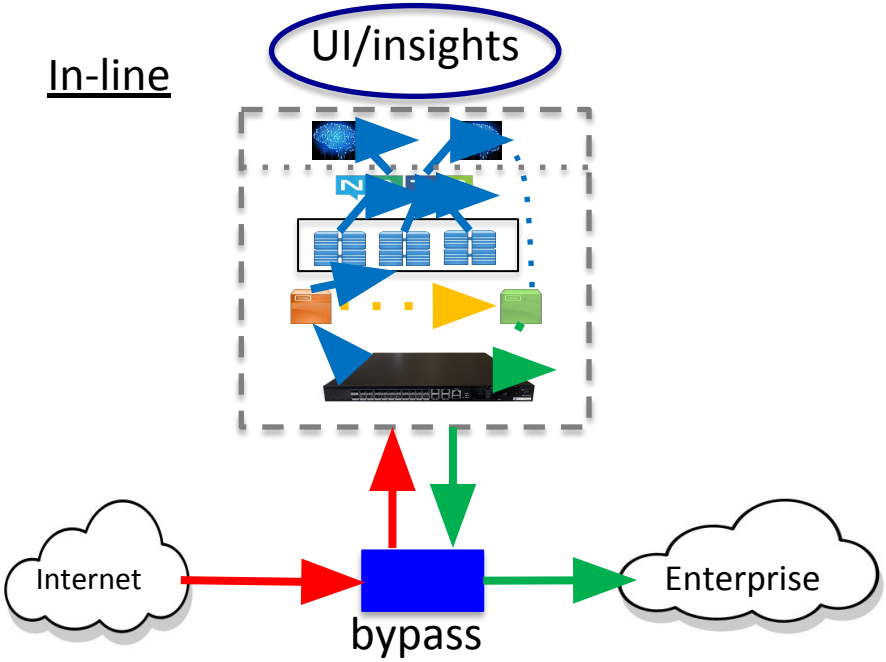




Time	Src IP	Dst IP	Packet Type	Queried Name	Anomaly Score	Domain
21:51:44	139.214.92.98	202.9.0.68	Query	nbsdk-baichuan.alicdn.com	0.4791	alicdn.com
21:51:44	139.214.92.98	202.9.0.68	Query	tunion-api.m.taobao.com	0.3807	taobao.com
21:51:44	139.214.92.98	202.9.0.68	Query	btrace.qq.com	0.3867	qq.com
		202.9.0.68	Query	ia.xuanwu88.com	0.3982	xuanwu88.com

Potential models of Trial





Conclusions

- SDN-Accelerated-NF has a strong value proposition
  - Commodity hardware → lower cost
  - Multi-vendor → choice of port speeds and densities
  - Modular and elastic → scalable in performance
  - Unbundled → more innovative and agile "state-of-art" software (ML/AI)

- Looking for operators with curiosity and appetite
  - Co-develop and test innovative solutions (journey/learnings are important)
  - Gain new insights that would not emerge otherwise
  - Develop skills in emerging areas (software/ML) in work-force
  - Please come talk to us <u>today</u> for a potential trial

Sponsors

- Our SDN Research has been supported by:
  - Australian Research Council (ARC)
  - Google
  - Optus, Telstra, AARNet
  - Cisco, HPE, NoviFlow
  - DSTG (Defence Sciences Technology Group)
  - Open Networking Foundation (ONF)